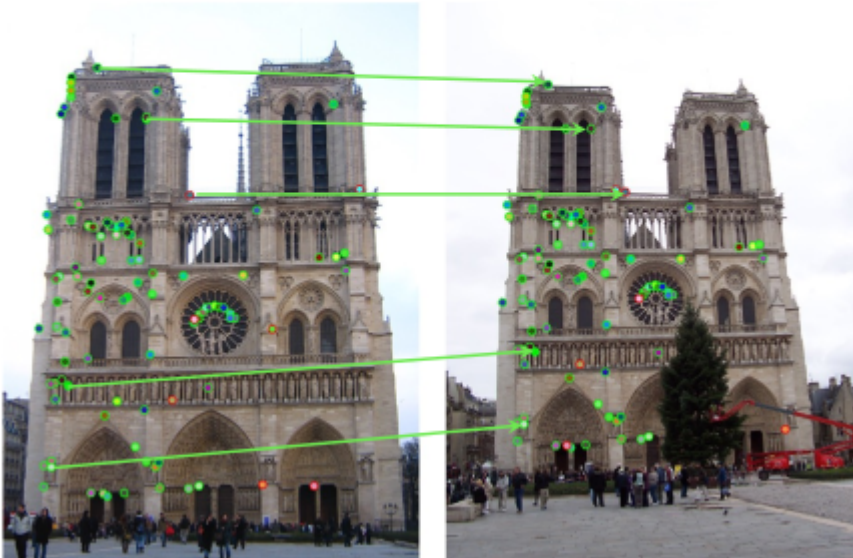


L12: Feature Descriptors

- Summarises local structure around keypoint

Can We Describe Keypoints to Enable Matching Across Images?



- Is all about the descriptor at keypoint
 - f

Popular Features Descriptors

- HOG: Histogram of Oriented Gradients
- **SIFT: Scale Invariant Feature Transform**
- SURF: Speeded-Up Robust Features
- GLOH: Gradient Location and Orientation Histogram
- **BRIEF: Binary Robust Independent Elementary Features**
- **ORB: Oriented FAST and rotated BRIEF**
- BRISK: Binary Robust Invariant Scalable Keypoints
- FREAK: Fast REtinA Keypoint

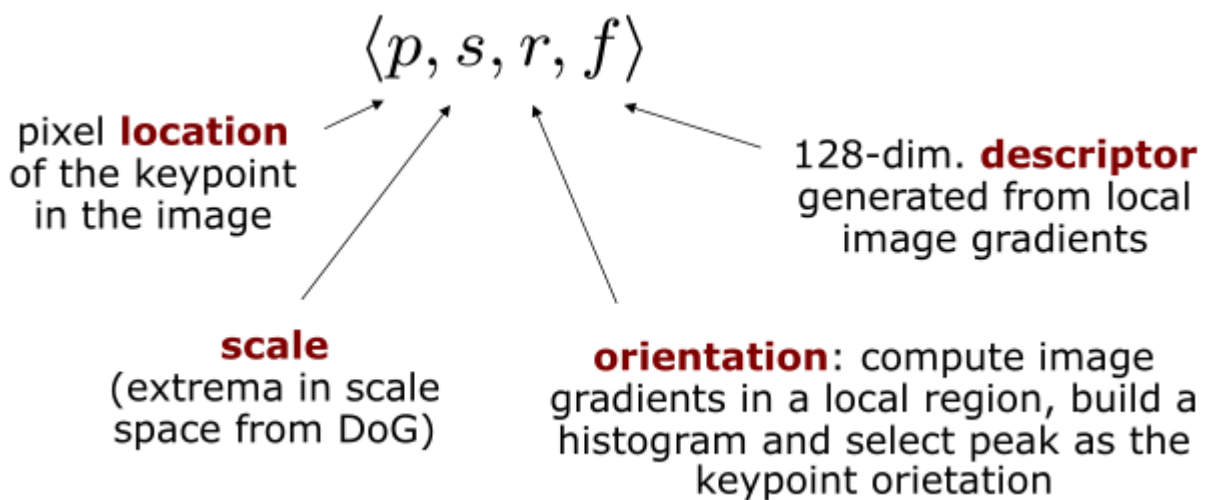
SIFT Descriptor

- Image transformed into features
- Invariant to:
 - Image translation

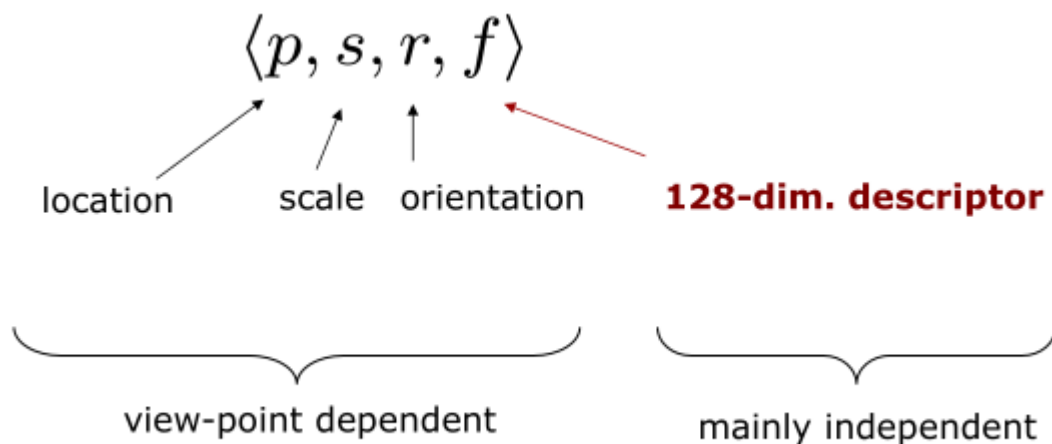
- rotation
- scale
- Partially invariant to
 - illumination changes and
 - affine transformations and 3D projections
- Suitable for detecting visual landmarks
 - from different angles and distances
 - with a different illumination

SIFT Features

- Given by vector
 - computed at local extreme point in scale space



◦

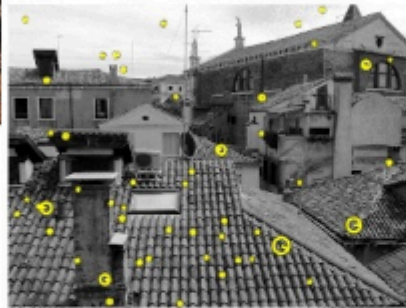


◦

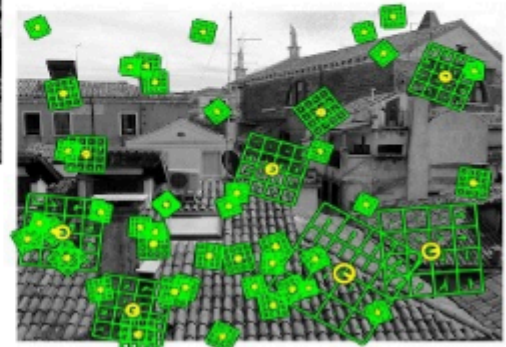
- SIFT considers distribution of gradients around keypoints
 - local 16×16 area at selected scale



input



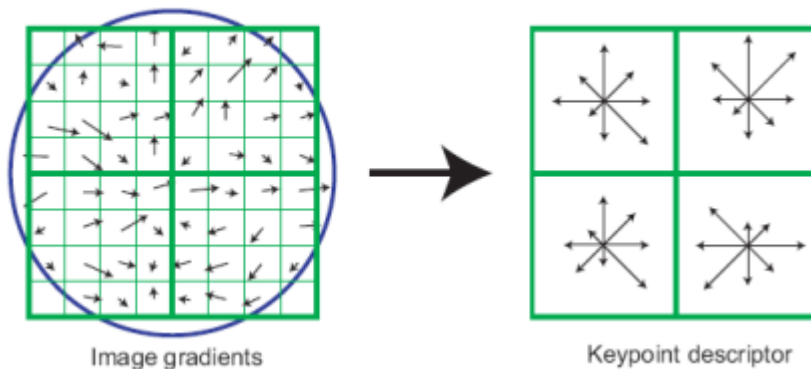
key points



descriptor (regions)

-
- Create array of orientation histogram
- $8 \text{ orientations} \times 4 \times 4 \text{ histogram array} = 128 \text{ dimensions}$
 - yields best results

Example using an 8x8 area:

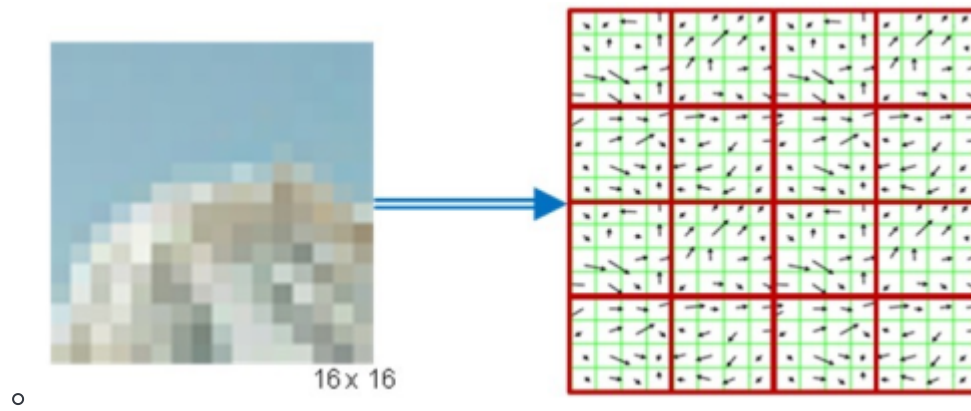


1. For given keypoint,
 - Warp the region around orientation
 - Scale it
 - resize to 16×16 pixels

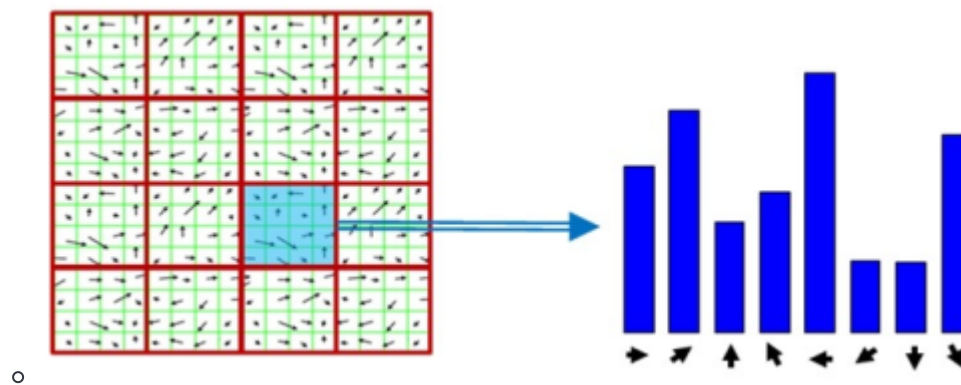


2. Compute gradients for each pixel

- orientation and magnitude
- Divide pixels into 16, 4×4 pixel squares



3. Compute gradient direction histogram over 8 directions



4. Concatenate histograms

- 128 dimensional feature vector



- Match on Descriptor Difference

Lowe's Ratio Test

- Test to eliminate ambiguous matches for a query feature q

1. Step: Find closest 2 descriptors to q , called p_1 and p_2

- based on Euclidian distance d

2. Step: $d(q, p_1) < T$

3. Step: Accept match only if best match is substantially better than second defined as:

- $\frac{d(q, p_1)}{d(q, p_2)} < \frac{1}{2}$

Outliers

- Lowe's Ratio test works well

- Few outliers remain
- Outliers require extra treatment

Binary Descriptors

- Computing descriptors fast
- No patenting issues
- Aim at generating small binary strings
 - easy to compute and compare

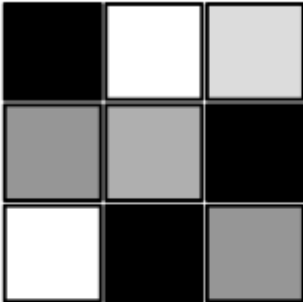
Strategy

- Select patch around a keypoint
- Select set of pixel pairs in that patch
- For each pair, compare intensities

$$b = \begin{cases} 1 & \text{if } I(s_1) < I(s_2) \\ 0 & \text{otherwise} \end{cases}$$

- Concatenate all b's to a bit string

Example

image area	index (for pairs)		
	1	2	3
	4	5	6
	7	8	9

pairs: $\{(5, 1), (5, 9), (4, 6), (8, 2), (3, 7)\}$

tests: $b = 0 \quad b = 0 \quad b = 0 \quad b = 1 \quad b = 1$

result: $B = 00011$

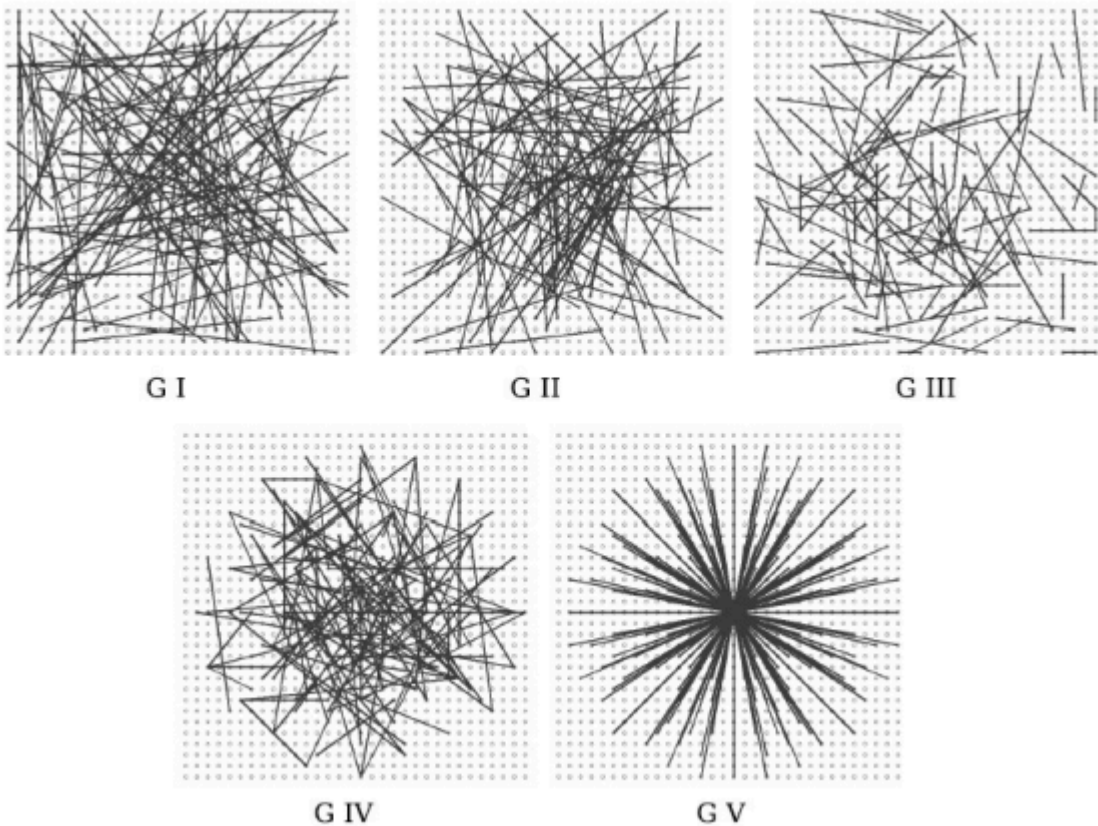
- Differ by the Strategy of selecting pairs
- To compare descriptors among images:
 - Use same pairs
 - Maintain order of testing

Advantages

- **Compact descriptor:** Number of considered pixel pairs defines the descriptor length in bits
- **Fast to compute:** Simply intensity value comparisons
- **Trivial and fast to compare:** Hamming distance

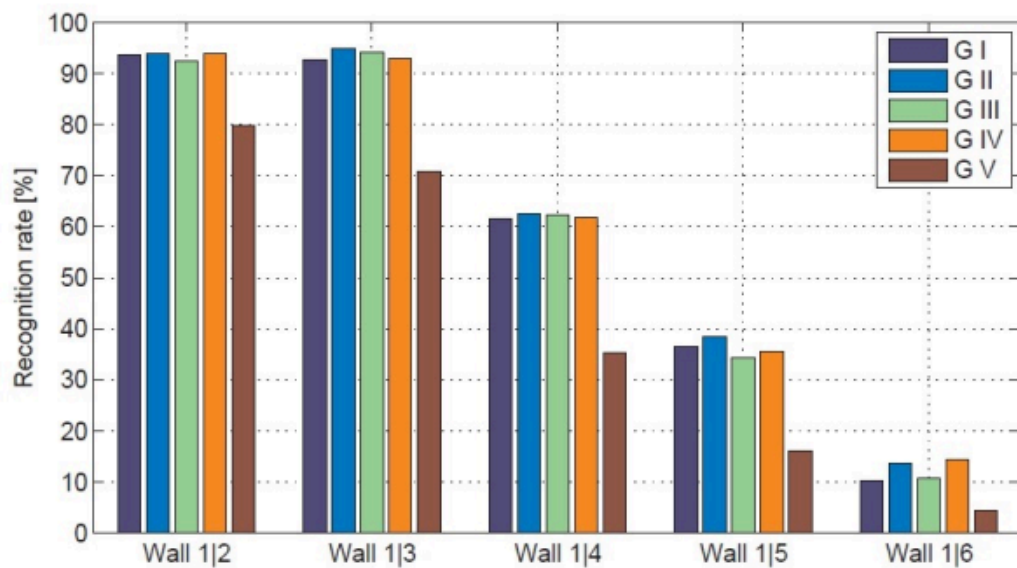
BRIEF: Binary Robust Independent Elementary Features

- First binary image descriptor
- 256-bit descriptor
- 5 different geometries as sampling strategies
- Noise: operation performed on a **smoothed** image to deal with noise



-
- G I: Uniform random sampling
- G II: Gaussian sampling
- G III: s_1 Gaussians; s_2 Gaussian centered around s_1
- G IV: Discrete location from a coarse polar grid
- G V: $s_1 = (0, 0)$; s_2 are all location from a coarse polar grid

Performance: G I – G IV are all good, G V less useful

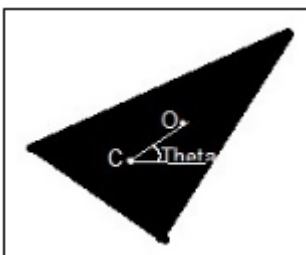


ORB: Oriented FAST Rotated BRIEF

- Extension to BRIEF:
 - Adds rotation compensation
 - Learns optimal sampling pairs

Rotation Compensation

- Estimates center of mass and main orientation of area/patch
- Image moment
 - $m_{pq} = \sum_{x,y} x^p y^q I(x,y)$
- Center of mass
 - $C = (\frac{m_{10}}{m_{00}}, \frac{m_{01}}{m_{00}})$
- Orientation
 - $\theta = \arctan 2(m_{01}, m_{10})$



- Given CoM and orientation C, θ ,
 - We can rotate coordinates of all pairs by θ around C:

- $s' = T(C, \theta)s$
- Use transformed pixel coordinates for performing test
- Invariance to rotation in the plane
- Pairs should be/have
 - **uncorrelated** - each pair adds information to descriptor
 - **high variance** - make feature more discriminative
- ORB defines strategy for
 - selecting 256 pairs
 - optimising for both properties using training database

ORB vs SIFT

- ORB 100x faster
- ORB: 256-bit vs SIFT: 4096-bit
- ORB: not scale-invariant
 - But achievable via image pyramid
- ORB mainly in-plane rotation invariant
- ORB(without Scale) similar matching performance
- Several modern online systems (e.g. SLAM) use binary features
- Gradient histograms:
 - SIFT, SURF
- Binary descriptors for efficiency
 - (BRIEF, ORB,...)